

I thank my whole family, from my grandparents and parents to my siblings, for all the support they have given me throughout my life, the opportunities they have provided me, and the sacrifices they have made.

I thank all my professors over the years who inspired me and shared their passion. In particular, Prof. Biggio for his guidance and feedback during these months.

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1 Introduction

1.1 Purpose

The purpose of this thesis is to explain how companies within manufacturing and chemistry are using Artificial Intelligence (AI) in their operations, products, and organizational capabilities. The first part of the work presents the applications described in academic studies, engineering papers, and other technical literature. These sources provide an overview of the key areas where AI has been implemented in the sector. However, they may overlook what a specific company is currently developing or trying to implement, especially in relation to the newest trends. Indeed, an academic study may show a technical possibility and an engineering paper may describe a specific experiment or collaboration, but they may not necessarily show whether a capability is being adopted inside the firm. For this reason, the second part of the thesis focuses on how fragmented public data can be processed to map current company activity, with the aim of combining several sources, such as annual reports, job postings, and public sources, to reconstruct where this technology appears inside each company and how these signals relate to industrial developments. Annual reports can provide an overview of the investments and of the board's direction, whereas job postings can show which skills and capabilities are being requested through recruitment, and public information can add details about internal programs. Together, they can provide a better view of where this type of activity is visible. Given this, this analysis is organized around a pipeline that converts job descriptions into structured reports, expands them with external data, evaluates their reliability, and indexes them in a RAG chatbot. Since the sources used in this study differ in the type of evidence they give, the hiring reports are treated as predictors of what the companies are building, while public details are exploited to support claims about deployment. Thus, the purpose is to collect what is known in the industry from the applications already described in the technical literature

and then move to the evidence that supports what individual firms are doing. Then, the final company reports can be queried through natural language, allowing the user to ask about individual companies, compare capability areas, or explore sector trends.

1.2 Sector Context

Manufacturing and chemistry have been chosen considering the impact that AI can have in the value chain of the companies in the sectors, from internal processes in terms of production, research, operations and business activities. In manufacturing, McKinsey describes AI as part of the Fourth Industrial Revolution (McKinsey & Company, 2024a), and estimates that generative AI could generate between \$2.6 trillion and \$4.4 trillion in annual value for the global economy (McKinsey & Company, 2023). Analytical AI can be applied to pricing, forecasting, process control, and production optimization. Generative systems can support molecular and material discovery or knowledge extraction from technical documents (McKinsey & Company, 2024b). Hence, AI is applicable to multiple phases in the chemical industry workflow. The global AI market in chemicals is projected to expand from \$943.0 million in 2023 to \$5.24 billion by 2030 (Grand View Research, 2024).

It can also impact how plants are monitored, how defects are detected, and how technical information is used in the company. Earlier forms of automation and predictive modelling were usually applied to specific tasks, while recent AI systems can process both structured data and technical language, allowing information from production, materials research, and business operations to be combined more smoothly than before. For this reason, newer systems based on LLMs, RAG, and agentic AI are especially relevant in these industries because they can connect information that is usually handled separately, from technical documents and laboratory records to plant workflows and sensor data. Considering this variety of use cases, manufacturing and chemistry are suitable for the objective of this thesis because artificial intelligence can appear in many parts of the firm, while external information is often fragmented. More specifically, the analysis focuses on three NACE sectors, 22.11, manufacture of rubber tyres and tubes, 22.19, manufacture of other rubber products, and 29.32, manufacture of other parts and accessories for motor vehicles. These were selected because they represent an interesting case for analyzing company activity from the outside.

1.3 Methodology

The methodology consists of a literature review and an empirical experiment, with the descriptive part based on a selected review of technical and academic sources. Since the objective was to explain the systems relevant for manufacturing and chemistry, surveys and review papers were used to identify the primary applications, including predictive maintenance, Retrieval Augmented Generation (RAG), and Agentic Artificial Intelligence.

Some publications were selected because they introduced foundational architectures, such as transformers, graph neural networks, or agentic tool use. Other papers were selected because they provided applied examples, figures, or benchmarks connected to manufacturing and chemistry. Finally, the timing of each publication was considered according to the technology discussed, in order to reflect the state of the art. For LLMs, RAG, and agentic AI, more recent works were preferred because they are developing quickly, while older papers were kept when they introduced models, methods, or benchmarks that are still used as the basis for the technologies discussed later.

In the experiment, the objective was to implement an LLM pipeline to process fragmented public data and produce final company reports. Considering the recent advances in the field, it was important to examine both their use cases and their underlying functioning, in order to understand what was feasible for this purpose. RAG and agentic AI were therefore treated in more detail because their benchmarks helped define the evaluation used for the final company reports, where each factual claim was assessed according to whether it was supported by the cited source in the documents later used by the chatbot.

2 Modern AI Systems

2.1 Background

Before the recent spread of LLM based systems, AI was already present in manufacturing and chemistry through various specialized tools built to address specific technical functions such as fault detection, predictive maintenance, quality assurance, materials discovery, reaction modelling, and digital twin optimization (Fahle *et al.*, 2020; Srinivasan *et al.*, 2025). These tools usually required the relevant process to be reduced to a machine readable representation, since they relied on specific inputs such as sensor histories, inspection images, molecular graphs, process diagrams, or other structured records (Srinivasan *et al.*, 2025; Chiang *et al.*, 2022). LLMs can instead work with technical language and therefore connect functions, data sources, and stages of work that had previously been handled separately. On this basis, newer architectures such as RAG and agentic AI emerged, linking documents, records, models, and tools within the same workflow, and, in the latter case, also supporting multi step task coordination. As production systems, plants, and equipment incorporated more sensors and monitoring devices, firms started to collect growing volumes of machine data and production images (Plathottam *et al.*, 2023; Ucar *et al.*, 2024). In this context, predictive maintenance and visual quality control became early industrial uses of machine learning because both helped firms decide when to intervene before failures or in case of defects, which could otherwise disrupt production and reduce product quality (Fahle *et al.*, 2020; Plathottam *et al.*, 2023; Chiang *et al.*, 2022). In visual quality control, convolutional neural networks were used to detect and classify defects directly from images, since they analyze local regions of an image, learn relevant visual features, and use them to distinguish acceptable products or surfaces from defective ones. (Fahle *et al.*, 2020; Plathottam *et al.*, 2023).



Figure 2.1: Example of a CNN for image based quality control. Elaboration based on: (Chiang *et al.*, 2022).

Instead of relying on visible deterioration, predictive maintenance aimed to detect earlier precursors in sensor signals by analyzing patterns distributed across multiple process variables (Chiang *et al.*, 2022; Ucar *et al.*, 2024). For this reason, PCA, and later t-SNE and UMAP, were used to reduce high dimensional measurement histories into lower dimensional representations where abnormal behaviour became easier to identify and compare (Chiang *et al.*, 2022).

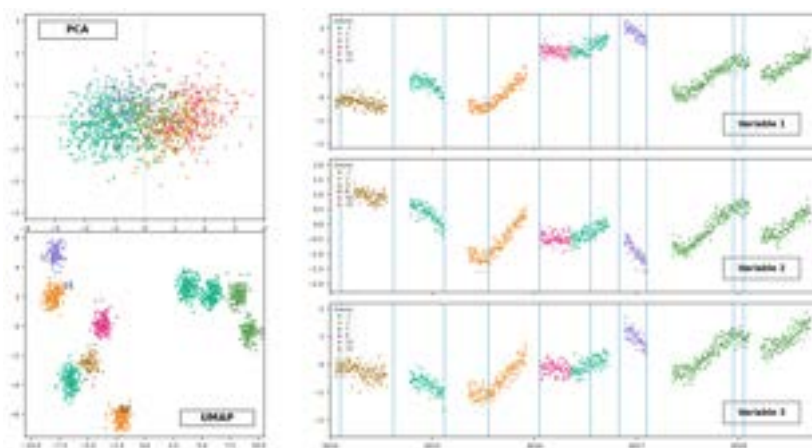


Figure 2.2: PCA and UMAP for process monitoring. Elaboration based on: (Chiang *et al.*, 2022).

These representations were used to separate normal operating regimes from irregular ones, and were then passed to more specific systems depending on the task. Classification methods such as support vector machines or random forests were used to classify sensor patterns and support fault detection or failure signature modelling, while LSTMs were used in sequence based settings where the objective was to track degradation over time and estimate remaining useful life (Chiang *et al.*, 2022; Ucar *et al.*, 2024).

Industrial use also extended to molecules, reactions, and materials, where prediction depended on chemical and structural representations. Earlier models usually relied on

intermediate representations that translated molecular structure into a simpler machine readable description, which could take the form of numerical features or compact strings such as SMILES.



Figure 2.3: String based molecular representations example. Source: (Nguyen-Vo *et al.*, 2024).

This reduced complex chemical structures to inputs that were easier to screen and relate to target properties across large sets of candidates. Graph based approaches were used because they preserved molecular structure instead of encoding it only as a fixed description, as some earlier representations often did (Srinivasan *et al.*, 2025). In graph neural networks, each atom was characterized through the neighbouring atoms connected to it, which allowed the model to capture local chemical environments across the structure (Andraju, 2025). This was useful both for estimating molecular and material properties and for narrowing the search for promising candidates before synthesis and experimental validation. Google DeepMind's GNoME is a well known example, since it used graph neural networks to predict more than 2.2 million crystal structures, around 381,000 of which were estimated to be computationally stable (Merchant *et al.*, 2023; Andraju, 2025).

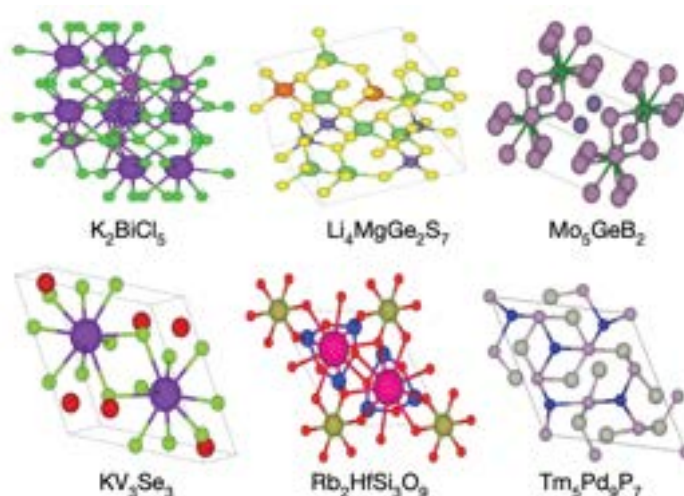


Figure 2.4: Examples of crystal structures identified by GNoME among the predicted candidate materials. Source: (Merchant *et al.*, 2023).

It was also extended to reaction outcome prediction, where representing the system as a graph preserved the relations between reactants and other key components, supporting the estimation of yield and side products (Andraju, 2025).

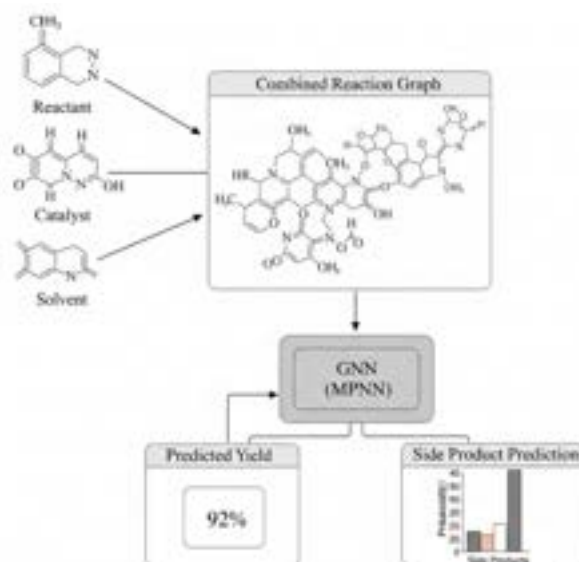


Figure 2.5: Graph neural network for reaction prediction, where reaction components are combined into a reaction graph to predict yield and side products. Source: (Andraju, 2025).

While the previous applications focused on predicting single chemical structures in the laboratory, AI was also employed at the process level in industrial plants to model how units such as reactors and process lines behaved during production (Srinivasan *et al.*, 2025). In these settings, the relevant structure was both molecular and physical, since the model had to capture how operating conditions, such as temperature and flow rate, evolved across the equipment and how these changes influenced reaction performance and product quality (Srinivasan *et al.*, 2025). One of the main approaches for these tasks was the hybrid model, since it combined an engineering description of how the plant was expected to behave with a predictive component learned from plant data (Srinivasan *et al.*, 2025; Langlotz *et al.*, 2022). The first part described how the system was expected to respond, while the second adapted that description to actual operating behaviour (Srinivasan *et al.*, 2025; Langlotz *et al.*, 2022). In practice, hybrid approaches were used to anticipate the effect of changing operating conditions before intervening on the plant, giving a more reliable basis for deciding whether to adjust the system and reduce undesired behaviour (Plathottam *et al.*, 2023; Langlotz *et al.*, 2022).

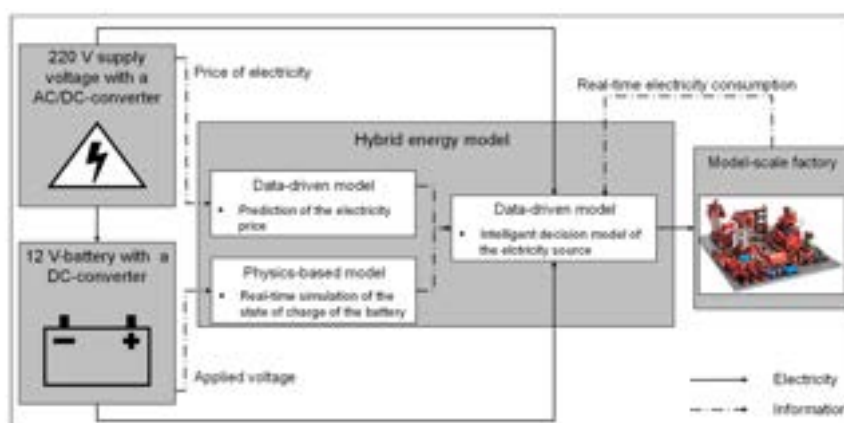


Figure 2.6: Hybrid model for industrial energy management, combining electricity prediction with battery simulation to support decisions on how to adjust plant operation before intervening on the factory. Source: (Langlotz *et al.*, 2022).

While hybrid models provided a predictive representation of the plant and estimated how the system would respond to possible adjustments, digital twins connected it to the actual production line through sensor updates and live data flows. This allowed potential adaptations to be tested in the virtual environment before being applied to the physical plant, while still being based on current operating data. Consequently, intervention decisions were more grounded in the observed behaviour of the plant, since changes could be assessed from actual data before being implemented in production (Srinivasan *et al.*, 2025; Andraju, 2025; Huang *et al.*, 2021).

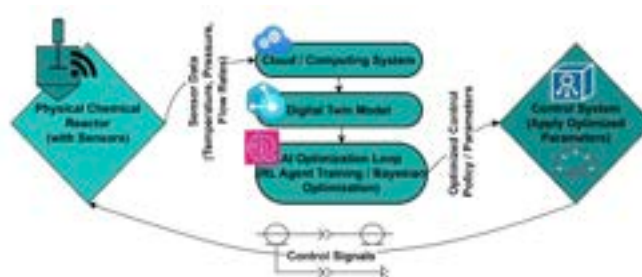


Figure 2.7: Digital twin for a chemical reactor, where sensor data from the plant are used to keep the digital model aligned with the system during production. Source: (Andraju, 2025).

Alongside the structured data sources discussed so far, critical information often remained embedded in scientific papers, patents, and technical documentation. For this reason, NLP started to be used to turn these unstructured texts into machine readable datasets. Concretely, this involved extracting chemical species and process variables, allowing researchers to reconstruct reaction procedures and feed them into the modelling pipelines discussed above (Chiang *et al.*, 2022; Andraju, 2025). Some

representative examples included ChemicalTagger for parsing chemical experimental documents, and tools such as IBM DeepSearch for structuring reaction information from papers and patents (Hawizy *et al.*, 2011; IBM Research, n.d.).

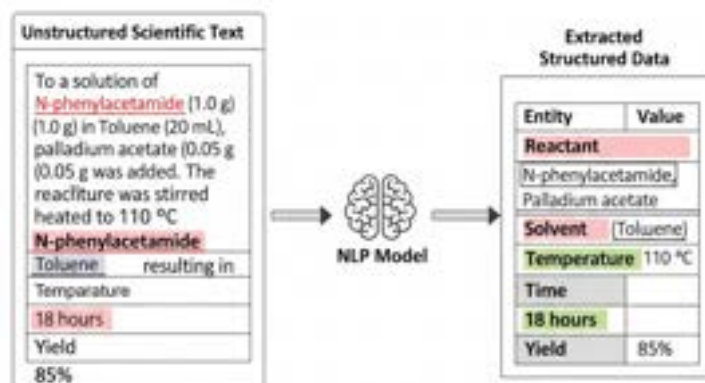


Figure 2.8: NLP pipeline for extracting structured reaction data from scientific text, where entities and process variables are identified in unstructured documents and converted into machine readable records. Source: (Andraju, 2025).

A further application was the extraction of negative data, such as failed reactions or unwanted byproducts. Unlike successful procedures, these outcomes were less systematically reported and usually remained hidden within the general text. By parsing these sentences, natural language processing algorithms could associate specific chemical species and experimental conditions with negative results. This allowed the predictive models discussed earlier to learn not only from successful cases, but also from the exact conditions that led to failure (Andraju, 2025).

As the previous examples show, the scope of these models was typically restricted to specific narrow applications. Since they were engineered for distinct data formats, they remained isolated within their function inside the pipelines. Integrating different plant operations was therefore difficult, as it required combining entirely different structures.

2.2 LLM

Traditional NLP systems treated classification and information extraction as distinct problems, since one model was trained to assign a label to an entire text, while another was tailored to locate within that same text the entities and process variables relevant for further analysis. A different approach emerged with the introduction of the transformer architecture, which replaced the recurrent and convolutional structures used in earlier sequence models with an attention based mechanism, therefore providing a more flexible basis for handling different language processing tasks (Vaswani *et al.*, 2017). Building on this foundation, transformer language models were trained on large unlabeled text corpora, allowing one model to learn complex linguistic and structural patterns (Radford *et al.*, 2018). This reduced the dependence on pipelines for specific tasks, because the same pretrained model could address multiple forms of text processing instead of training separate models for each individual application. Within integrated industrial operations, LLMs advanced from basic information extraction to enabling more active forms of decision support. By functioning as conversational interfaces, they could integrate safety guidance directly with equipment documentation, allowing operators to query regulations and technical instructions more directly (Singh *et al.*, 2025). In parallel, they also extended research workflows in chemistry and materials science by supporting computational simulations, molecular and material design, and tasks related to materials discovery (Zimmermann *et al.*, 2025; Li *et al.*, 2024). When connected to external systems, the LLM could also serve as a reasoning and coordination layer, creating the basis for more agentic workflows in industrial and research settings (Plaat *et al.*, 2025; Zimmermann *et al.*, 2025). As a practical example, conversational interfaces can be deployed in shop floor settings to integrate safety management and technical support into manufacturing operations. A recent study illustrated this by proposing a multimodal chatbot that allows workers and operators to troubleshoot equipment by verbally asking questions or uploading images of the plant machinery. The main strength of this approach is its ability to process technical documentation across entirely different types of industrial machinery, effectively making

multiple technical sources available through one tool. By delivering responses in the evaluated prototype with nearly 87 percent accuracy and an average response time of approximately ten seconds, the system could help workers clarify procedures and safety related doubts more quickly, making operating instructions more accessible within the physical constraints of the plant (Singh *et al.*, 2025).

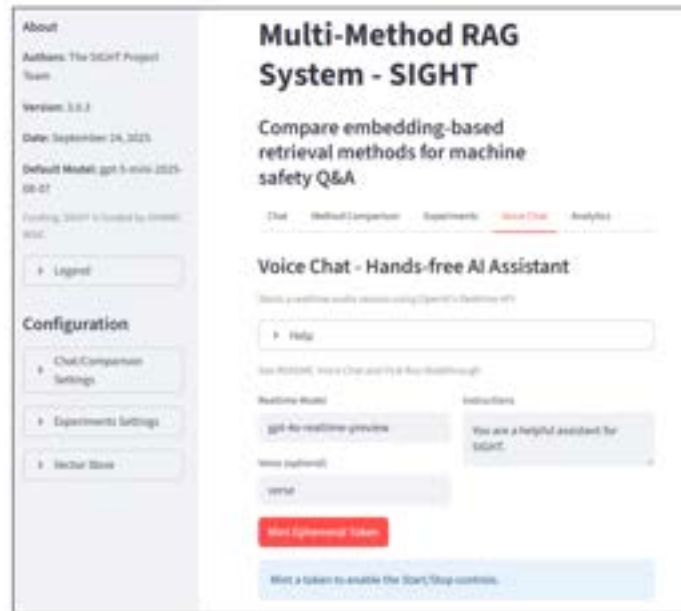


Figure 2.9: Multimodal manufacturing safety chatbot interface for operator support. Source: (Singh *et al.*, 2025).

Many manufacturing activities depend on software tools with their own syntax and commands, such as CAD software for designing components, CAM software for machining steps, and simulation tools for material stress. In these cases, engineers could describe the task in natural language and provide relevant specifications or design files, while the model translates that request into the script or command sequence required by the specific software. A practical example is the design of manufacturable components, such as brackets, frames, or plates. Instead of building the geometry manually from the beginning, the model can generate a first three dimensional prototype from the required shape, dimensions, and functional constraints provided as input. Engineers would still need to check the result and adapt it to production requirements, but the initial setup would require less manual work than creating the component entirely from scratch in the design software (Zimmermann *et al.*, 2025; Li *et al.*, 2024; Makatura *et al.*, 2024). At the R&D stage, one of the main objectives is to identify which material or formulation is most



Figure 2.10: Text to CAD example. Source: (Li *et al.*, 2024).

promising for a target property, such as thermal stability or mechanical performance, before investing in laboratory experiments or further validation. In particular, a candidate material can be described through its composition or molecular structure, both of which can be expressed in textual or structured formats. From these descriptions, the model could learn how certain structural or compositional patterns were associated with measured or simulated properties, and then use this relation to estimate how a new candidate might behave or suggest variations to be tested next. After a target property has been defined, the system can help identify candidate materials worth ranking, modifying, and testing. The result is a list of candidate materials to check, still requiring simulations, experiments, and expert validation (Zimmermann *et al.*, 2025; Miret and Krishnan, 2025). Across these applications, the main constraint was related to keeping information up to date, so that the system could answer according to current safety regulations, the syntax of recently introduced programming languages, or newly discovered materials. In these cases, the pretrained model alone was not enough, since the answer had to be tied to the sources available at the moment of use. For this reason, retrieval augmented generation (RAG) connects the model to external documents and records before generating the response, so that its output can reflect domain specific and up to date information.

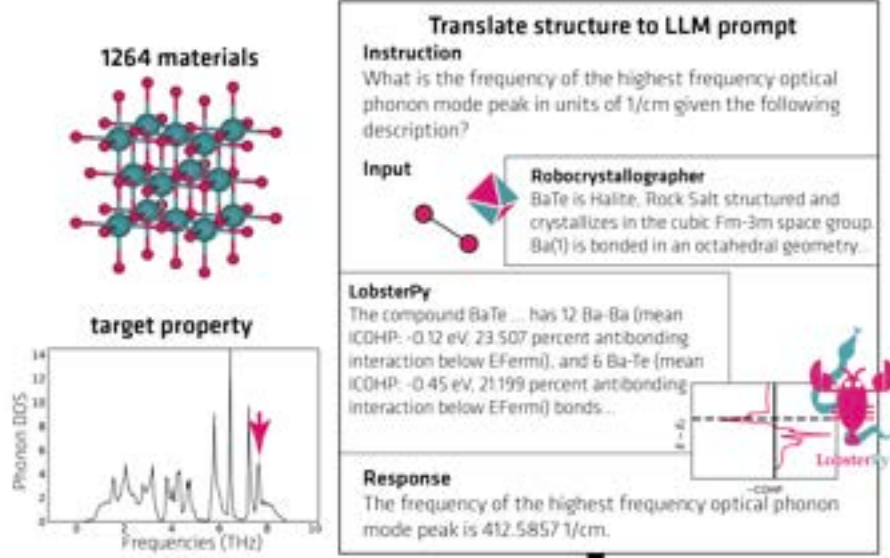


Figure 2.11: LLM based material property prediction, where crystal structures are translated into textual descriptions used to estimate target material properties. Source: (Zimmermann *et al.*, 2025).

2.3 RAG

Retrieval Augmented Generation (RAG) reduces the dependence of foundation models on knowledge stored in their parameters during training, as this information is difficult to inspect, revise, or update after deployment, leading to plausible answers that are not supported by a verifiable source. For this reason, RAG adds a non parametric memory, in the form of an external collection of documents that the system can search when a query is submitted. The response is still generated by the LLM, but its content is conditioned on sources that can be updated, replaced, and inspected (Lewis *et al.*, 2020; Guu *et al.*, 2020; Gao *et al.*, 2023).

The system indexes external memory to retrieve relevant document segments for each user query. Let $D = \{z_1, \dots, z_n\}$ represent the external document collection, where each z_i is a text chunk from sources such as manuals or scientific papers. Both the user query x and each chunk z_i are converted into vector representations. The retriever assesses the relevance of each chunk to the query, typically using an inner product $s(x, z_i) = q(x)^\top d(z_i)$, where $q(x)$ and $d(z_i)$ are respectively the vector representations of the query and document chunks. The system selects the K highest-scoring chunks, $Z_K(x) = \text{topK}_{z_i \in D} s(x, z_i)$, and adds them to the prompt as external context for answer generation. Each retrieved segment is treated as a latent variable, since the system

does not know in advance which will support the correct answer, and the probability of generating an answer y is approximated as $p(y|x) \approx \sum_{z \in Z_K(x)} p_\eta(z|x) p_\theta(y|x, z)$, where $p_\eta(z|x)$ is the relevance assigned by the retriever and $p_\theta(y|x, z)$ is the probability assigned by the generator when using both the query and the retrieved segment. The quality of the final answer depends on the entire pipeline since the retriever must select the relevant evidence, the augmentation must place it in the model context, and the generator must use it without introducing unsupported information (Lewis *et al.*, 2020; Gao *et al.*, 2023).

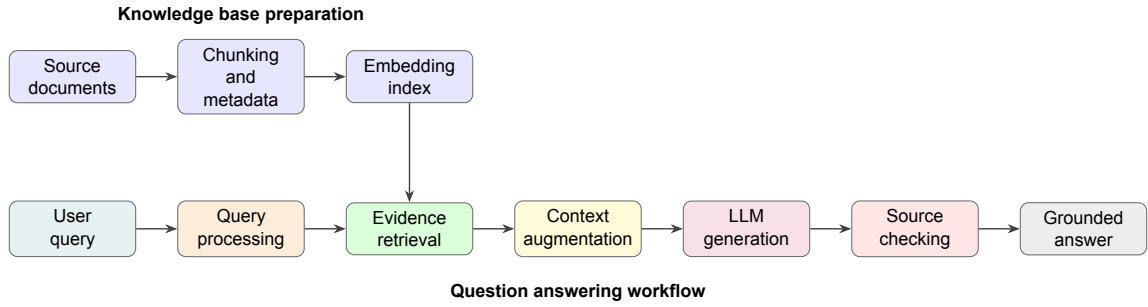


Figure 2.12: RAG pipeline example.

The primary evaluation concerns the retrieval stage and asks whether the top K retrieved segments contain the evidence required by the query. Let $R_K(x)$ be the set of the top K document segments retrieved for a query x , and let $E(x)$ be the set of segments that actually include the relevant evidence. Hit@K , $\text{Hit@K}(x) = \mathbf{1}\{R_K(x) \cap E(x) \neq \emptyset\}$, checks whether at least one relevant segment is in the retrieved set. Recall@K measures how much of the relevant evidence was retrieved, $\text{Recall@K} = \frac{|R_K(x) \cap E(x)|}{|E(x)|}$. Mean Reciprocal Rank instead considers the position of the first relevant segment, $\text{MRR} = \frac{1}{r}$, where r is its rank in the retrieved list. Together, these measures show whether the retrieved context contains the necessary evidence, whether important pieces are missing, and whether the most useful segments appear near the top of the retrieval list (Gao *et al.*, 2023).

The role of the benchmarks is to apply these measures to a defined set of questions, documents, and expected outputs, testing different RAG systems on the same queries and document collection, and comparing them on both retrieval and generation. Each example is therefore described through the user query q , the retrieved context $c(q)$, and the generated answer $a(q)$, so that $c(q)$ can be evaluated by checking whether

it contains the information needed to answer q and whether $a(q)$ uses correctly that information (Es *et al.*, 2024; Saad-Falcon *et al.*, 2024; Chen *et al.*, 2024).

One example is RAGAS, which evaluates RAG outputs without relying on reference answers by separating context relevance, answer relevance, and faithfulness. Given the query and retrieved context, an evaluator judges which sentences contribute to the answer and distinguishes them from irrelevant ones. If only part of the context is useful, the score is lower because the retriever has added unnecessary information. If none of the sentences provide evidence, the retrieved context cannot support the query. Answer relevance tests whether the answer is appropriate to the query. If it has a low score, it means the model added noisy information or did not properly answer the query. Faithfulness compares the answer with the retrieved context by turning the response into factual statements and checking whether each one is supported by $c(q)$. If some statements are unsupported, the model has added information beyond what was provided in the retrieved context (Es *et al.*, 2024).

Another common benchmark is ARES, which creates labeled examples from the document corpus and uses them to train smaller judge models. A positive example is a triplet that satisfies a specific evaluation task, for instance because the passage is relevant to the query or the answer is supported by that passage. Negative examples are created by changing one part of the triplet, for example by replacing the passage with an unrelated one, using an answer taken from another context, or generating a contradictory answer. The trained judges are then used to score new RAG outputs on context relevance, answer relevance, and faithfulness. Since they are trained on synthetic data, ARES also uses a smaller set of human annotations to correct possible bias and produce confidence intervals (Saad-Falcon *et al.*, 2024).

While RAGAS and ARES evaluate whether the retrieved context and the generated answer are aligned, RGB focuses on cases where retrieval itself can introduce errors by controlling the context given to the model before generation. In practice, each query is associated with a gold answer and a controlled set of external documents. The model receives the query together with these documents, and its response is then compared with the expected behaviour for that test case. When the context includes passages

related to the topic but not containing the answer, the response is correct only if it still matches the gold answer. When the necessary evidence is removed, the expected output is a refusal or an indication that the available sources are insufficient. When the evidence is split across documents, the answer is checked for whether it combines those pieces correctly. When wrong facts are inserted into the context, the benchmark checks whether the model detects and corrects the factual error, or whether it is misled by the incorrect passage. RGB therefore helps distinguish failures caused by noisy retrieval, missing evidence, fragmented evidence, or misleading context (Chen *et al.*, 2024).

In manufacturing settings, foundation LLMs are limited by their reliance on general knowledge acquired during pretraining, which often lacks the specificity needed for operational decisions. RAG allows users to interact through a conversational interface while ensuring that answers are derived from targeted sources, such as machine manuals, technical documentation, or safety regulations, a requirement that appears clearly in the manufacturing safety chatbot discussed in the previous section. A chat assistant deployed in the plant is useful only if it provides the correct instructions for the machine and the procedures involved, since a generic answer may be unsafe even when it seems truthful. By using RAG, the operator's query is linked to the indexed manual, regulation, or safety document that applies to that case, and the answer is then grounded in that material. Moreover, when machines, manuals, or safety procedures change, the knowledge base can be revised by updating the indexed documents (Singh *et al.*, 2025).

Another example involves the generation of standard operating procedures (SOPs) in chemical laboratories. An SOP specifies how a laboratory operation has to be performed, from instrument setup and operating steps to quality control. If the procedure is incomplete or inconsistent, the result may not be reproducible and the operation may no longer comply with safety or quality requirements. In this application, RAG is used to generate SOPs because relying only on the model's pretrained general chemical knowledge would make it harder to keep the document tied to the specific



(a) ARES

inclusion of redundant information. To estimate context relevance, given a question q and its context $c(q)$, the LLM extracts a subset of sentences, S_{ext} , from $c(q)$ that are crucial to answer q , using the following prompt:

Please extract relevant sentences from the provided context that can potentially help answer the following question. If no relevant sentences are found, or if you believe the question cannot be answered from the given context, return the phrase "Insufficient Information". While extracting candidate sentences you're not allowed to make any changes to sentences from given context.

The context relevance score is then computed as:

$$CR = \frac{\text{number of extracted sentences}}{\text{total number of sentences in } c(q)} \quad (2)$$

(b) RAGAS

Figure 2.13: Examples of RAG evaluation benchmarks. Sources: Saad-Falcon et al. (2024) (a) and Es et al. (2024) (b).

equipment, material, and safety context. The user specifies the instrument, its brand and model, the material type, and the sections that should be included, and these inputs are then connected with retrieved manuals, regulatory guidelines, and scientific references. In practice, a new SOP is produced by combining the selected structure of the procedure with the technical details found in the relevant sources. For instance, the requested sections may define the order and content of the document, while an equipment manual may provide the calibration steps and instrument instructions to be inserted into those sections. In this way, the generated SOP is adapted to the new use case while remaining tied to verified laboratory material. The analyzed paper aims to mitigate hallucinations in this way, because the model is constrained by retrieved manuals, guidelines, and reference documents. However, the generated SOP should still be treated as a draft requiring expert review before practical use, especially for correctness, completeness, compliance, and practical usability (Mallouhy *et al.*, 2026).

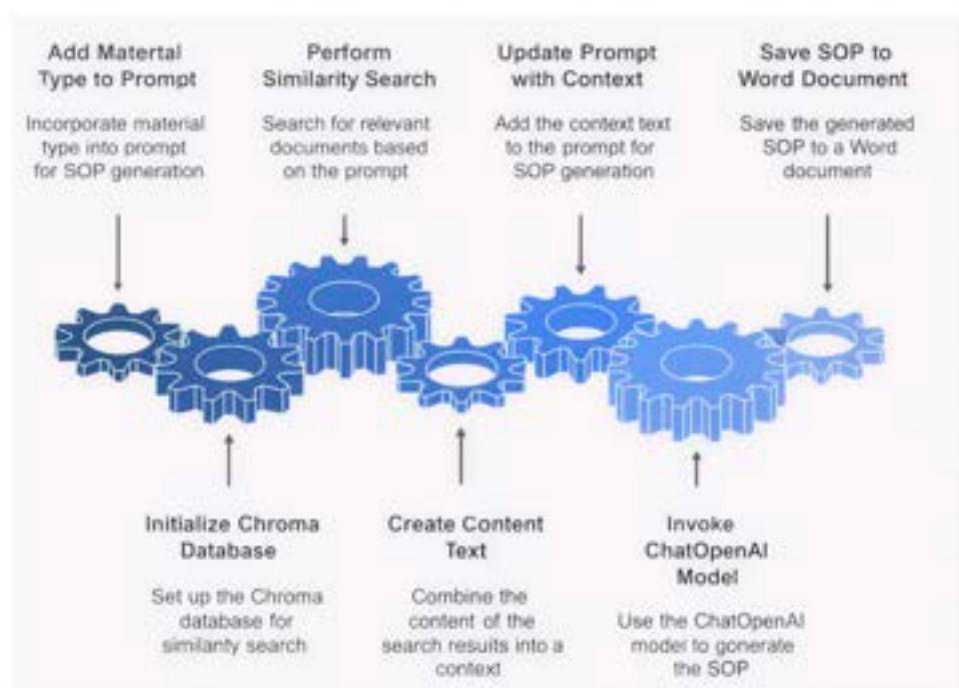


Figure 2.14: Workflow of the SOP generation process. Source: (Mallouhy *et al.*, 2026).

A third example is related to material design, in particular the search for metal organic frameworks with a lower band gap. Metal organic frameworks (MOFs) are crystalline porous materials formed by connecting metal units with organic molecules (Kumar and Kulkarni, 2023). Because their structure is modular, changing the metal unit or the organic linker can change the final material properties. In this case, the property of

interest is the band gap, which indicates how much energy is needed for the material to move electrons from a non conducting state to a conducting state. A lower band gap means that the material can interact more easily with visible light and conduct charge more effectively, which is useful in applications where the material has to absorb light, transfer electrons, or support catalytic reactions. RAG is used to modify an initial MOF structure according to design rules found in the scientific literature. The user provides an initial MOF as a SMILES string and states the property to improve, namely the reduction of the band gap. Since the band gap is a numerical property, the system can compare the starting material with each new candidate. For example, if the initial MOF has a predicted band gap of 2.3 eV, the objective is to suggest a chemically valid modification with a lower predicted value, such as 1.9 eV. In this case, the retrieved papers are used as sources of chemical strategies for lowering the band gap. For example, the system can extract suggestions such as increasing conjugation in the organic linker, using more electron rich metal nodes, or adding functional groups that affect electronic structure. These rules are then used to modify the initial MOF and produce a new candidate in SMILES format, which is checked with RDKit, a cheminformatics toolkit, because the generated string may describe an invalid or chemically unrealistic structure. If the candidate passes this check, a surrogate predictor estimates its band gap and uncertainty. Therefore, the new material can be compared with the starting MOF and with the previous candidates (Zimmermann *et al.*, 2025).

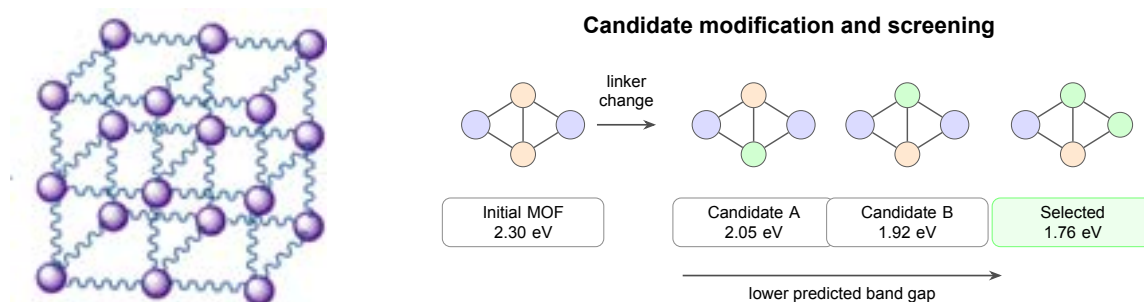


Figure 2.15: Example of a metal organic framework structure and schematic candidate modifications for lowering the predicted band gap. Source for the MOF structure: (Zimmermann *et al.*, 2025).

In more complex workflows, the LLM coordinates the steps required to complete the task, deciding when information has to be retrieved and when a tool has to be used. The

agent may use external knowledge to choose the correct tool, provide the right inputs, or interpret the result of an intermediate step. This knowledge can then support the next action in the workflow, with agentic AI extending RAG from grounded generation to multi step execution (Plaat *et al.*, 2025; Zimmermann *et al.*, 2025; Bran *et al.*, 2024).

2.4 Agentic AI

Agentic AI defines systems in which the LLM coordinates a task over several steps, acting as the decision maker of the agent. At each step t , the LLM receives the requested task g together with the available context, which includes the current state of the environment s_t and the memory of previous operations m_t , such as retrieved documents, tool outputs, or actions already performed. Based on this context, the model decides the next operation, which may involve querying a database, calling a software tool, running a simulation, or sending an instruction to an external robot. The operation is executed outside the LLM, and its result is returned to the model, which uses it to decide whether the task has been completed or whether another step is required. The next operation can therefore be represented in simplified form as $a_t = \pi_\theta(g, s_t, m_t)$. Once a_t is executed, the external system returns an updated state s_{t+1} , which becomes part of the context for the following step. Thus, the task is carried out through a sequence of decisions, operations, and observations (Plaat *et al.*, 2025; Wang *et al.*, 2024).

For this reason, the agent uses memory to store previous operations, retrieved documents, and tool outputs from earlier attempts. When the task is too complex to be completed in one action, planning decomposes it into intermediate steps and defines the order in which they should be attempted. Executing these steps requires a mechanism that allows the agent to identify the appropriate tool and provide the required inputs. Protocols such as the Model Context Protocol (MCP) support this mechanism by exposing each tool with a name, a description of its function, and a schema of the inputs it requires. These descriptions are included in the context available to the LLM, allowing the model to match the natural language request with the tool that can perform the required operation. The MCP server executes this call on the external system and returns the output to the agent, making it part of the context for the next step, so that the agent can continue the workflow, correct the previous operation, or stop when the task has been completed (Plaat *et al.*, 2025; Wang *et al.*, 2024; Anthropic, 2024).

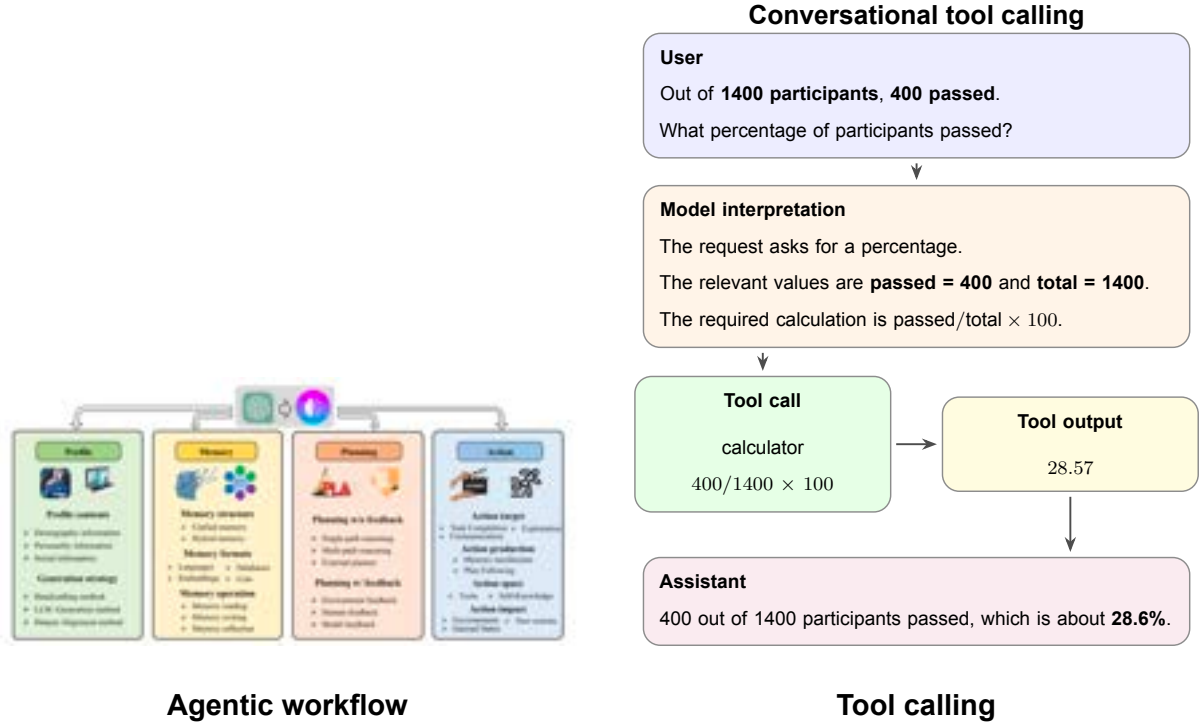


Figure 2.16: Examples of agentic workflow and tool calling. Workflow source: (Wang *et al.*, 2024).

To execute the selected operation, the model generates a structured call $c = (a_c, i_c)$, where a_c is the name of the API and i_c is the input passed to it. The external API processes this request and returns a result r , which the agent incorporates into its context for the next step. While protocols define how tool calls are exchanged with external systems, the model must still decide when a tool is actually needed. Instead of relying on large amounts of human annotations or task specific prompts to trigger these operations, Toolformer trains the model to decide tool use autonomously through self supervised dataset generation. Specifically, given an input text sequence $x = (x_1, \dots, x_n)$, the model evaluates possible tool interactions by sampling multiple structured API calls at different candidate positions and executing them externally to obtain the corresponding responses. After executing the candidate calls, Toolformer selects only those that reduce the loss on the following tokens. It compares $L_i^+ = L_i(e(c_i, r_i))$, where the model receives the API call c_i and its response r_i , with $L_i^- = \min(L_i(\epsilon), L_i(e(c_i, \epsilon)))$, where the model receives no call or only the call without the response. The call is kept only if $L_i^- - L_i^+ \geq \tau_f$, meaning that it improves the prediction of the next tokens. The selected calls are then inserted into the original text to create

the dataset used to fine tune the model. Toolformer applies this process to tools such as a calculator, a question answering system, a Wikipedia search engine, a machine translation system, and a calendar (Schick *et al.*, 2023).

A multi step task also requires the model to evaluate the result of each operation before deciding whether to continue, revise the plan, or stop. ReAct combines language reasoning steps with executable actions in a single task sequence, where reasoning is used to plan, track progress, and revise the next action. The sequence is updated during execution, since each observation returned by the environment becomes part of the context used to choose the following step. At step t , the agent receives an observation o_t from the environment and uses the previous trajectory as context, $c_t = (o_1, a_1, \dots, o_{t-1}, a_{t-1}, o_t)$. The model then selects the next element from an expanded action space $\hat{A} = A \cup L$, where A is the set of executable actions and L is the space of language reasoning steps. If the selected element belongs to L , the model produces a reasoning trace that updates the internal context by decomposing the task, tracking what has already been done, or adjusting the plan after an unexpected result, without changing the external environment. If the selected element belongs to A , the model performs an executable action. The environment then returns a new observation, which is added to the trajectory and used to choose the next decision. During the task, reasoning and action inform each other, since reasoning helps the model decide what to search, which action to take, or when enough information has been collected. The observations returned by the environment then show whether that reasoning was useful or whether the next step has to change. This reduces errors caused by relying only on the model’s internal knowledge and makes the process easier to inspect, because the final result can be traced through the intermediate thoughts, actions, and observations that produced it (Yao *et al.*, 2023).



Figure 2.17: Example of ReAct. Source: (Yao *et al.*, 2023).

In robotics, the same agentic structure is applied to physical space, where camera images and sensor observations are combined with the natural language instruction to decide the next operation. For example, PaLM-E produces high level instructions, meaning instructions that define what the robot should do, such as finding an object, grasping it, or moving it to another location. These instructions are then passed to lower level robot policies, which handle the physical execution by translating the task level instruction into robot actions and motor commands. Therefore, the LLM can reason about the task and the surrounding environment, but it does not directly control the robot motors. If o_t contains the current image or sensor observation and g is the task instruction, the model can produce a high level action, in simplified form, $a_t = \pi_\theta(g, o_t, m_t)$. A low level policy then maps this action into motor commands, $u_t = \mu(a_t, o_t)$. After the movement is executed, the physical environment changes and produces a new observation o_{t+1} , which is used to complete the task (Plaat *et al.*, 2025; Driess *et al.*, 2023).

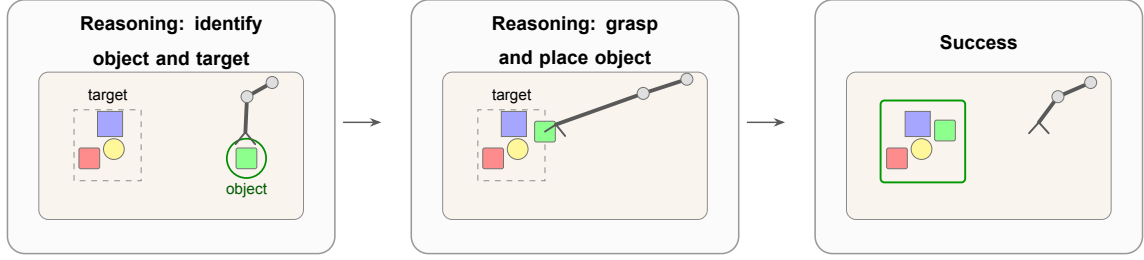
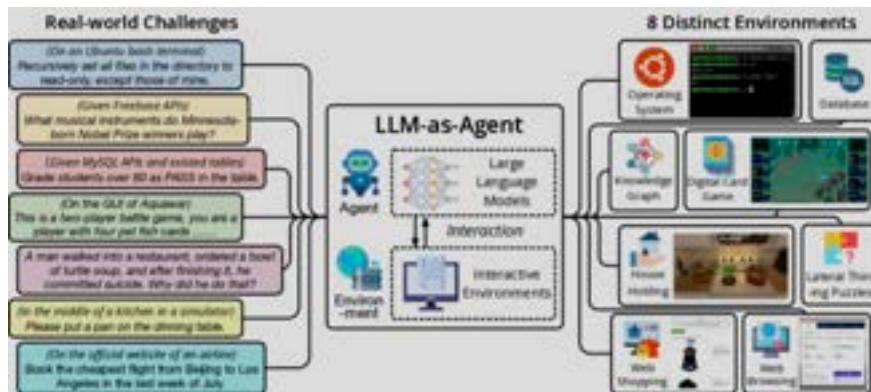


Figure 2.18: Language guided robot workflow. Conceptually derived from Driess et al. (2023).

For agentic AI, evaluation depends on whether the agent can transform a request into a sequence of steps, use external tools when needed, and return a verifiable result. At the benchmark level, GAIA evaluates this through real world questions that require combinations of reasoning, web browsing, file handling, multimodal understanding, and tool use. Each example is described as (q_i, y_i) , where q_i is the task given to the assistant and y_i is the expected final answer. During execution, the agent may produce an internal trajectory $\tau_i = (a_1, o_1, \dots, a_T, o_T)$, made of tool calls, observations, and intermediate steps, but the benchmark evaluates the final output $\hat{y}_i$. The score is computed as $\text{Acc} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{\text{norm}(\hat{y}_i) = \text{norm}(y_i)\}$, where $\text{norm}(\cdot)$ standardizes the model output and the ground truth before the evaluation applies quasi exact matching. Since the questions are designed so that the answer usually requires several operations, a correct final output indicates that the agent selected the necessary steps, used the required sources or tools, and combined their results correctly (Mialon *et al.*, 2023).

Another important benchmark is AgentBench, which moves evaluation beyond final answer matching toward interaction with an environment. The agent receives a task and has to produce valid commands over several turns, using the action format required by that environment. After each command, the environment returns feedback, such as a database result, a system output, a web page state, or a game observation, and the agent has to use that feedback to decide the next command. In particular, the benchmark includes eight environments grouped into code based, game based, and web based tasks, including operating systems, databases, knowledge graphs, a digital card game, lateral thinking puzzles, house holding, web shopping, and web browsing. The score is environment specific, because each setting defines success through its own final state, answer, or reward. For example, a database task is correct only if the

agent issues SQL commands that lead to the expected answer or table state, while a house holding task is correct only if the sequence of actions brings the simulated environment to the requested state. In this setting, the result depends on the sequence of commands used to reach it because an agent can understand the task and still fail during execution if it uses the wrong command format, passes incorrect arguments, loses context after several turns, or ignores the feedback returned by the environment. AgentBench is therefore useful when the objective is to evaluate the execution process itself, especially in tasks where the agent must interact with an environment until the required state is reached (Liu *et al.*, 2024).



AgentBench

Figure 2.19: AgentBench interactive environments. Source: (Liu *et al.*, 2024).

The manufacturing assistant discussed in the LLM and RAG sections could be extended into an agentic system when it is connected to production data and diagnostic tools. In the previous examples, the model supported operators by answering questions from manuals, safety documents, or machine specific procedures. Now, the same interface could also react to events from the plant, such as abnormal sensor readings, higher energy consumption, or a deviation in product quality. When such an event occurs, the agent would decide which checks are needed to understand the problem such as comparing the current signal with its historical trend, calling a predictive maintenance model, or testing a possible adjustment through a digital twin before any intervention is suggested. Then, the output would provide a recommendation for the operator or the control system, together with the documents, data, or model outputs that support it. For evaluation, the impact of the anomaly can be represented through a weighted cost measure, $J = w_1T_d + w_2D + w_3S + w_4E + w_5F$, where T_d is the time needed to

diagnose the anomaly, D is downtime, S is scrap rate, E is energy consumption per unit produced, and F is the number of false alarms. A lower value of J , compared with the same process managed without the agent, would indicate that the system has supported a faster and less costly intervention. Thus, this proposal shows how agentic AI could be applied directly inside production plants, helping human operators intervene more efficiently (Andraju, 2025; Plaat *et al.*, 2025).

A second application is the self driving laboratory, where the agentic loop is applied to experimental chemistry and materials synthesis. In this case, the environment is a laboratory platform that prepares samples, runs experiments, and uses the new data to decide the next experiment. The objective can be to improve the activity of a catalyst, increase the yield of a reaction, or optimize several properties at the same time. The agent selects an experimental condition x_t , which may include temperature, solvent, or concentration. Then, the laboratory executes the experiment and returns a measured result y_t , such as yield, stability, or catalytic activity. If the task involves several objectives, y_t can be treated as a vector containing the measured values for each property. The experimental memory is then updated as $D_t = \{(x_1, y_1), \dots, (x_t, y_t)\}$, and the next experiment is selected from this updated dataset. In this case, the system chooses the next condition by balancing the expected improvement in the target property with the uncertainty of the model. $x_{t+1} = \arg \max_{x \in X} \alpha(x; D_t)$, where $\alpha(x; D_t)$ gives a score to each possible experiment based on the expected improvement, the remaining uncertainty, and, when several properties are optimized together, the balance between activity, stability, and yield. The evaluation of the experimental campaign can then be measured by how quickly the system approaches the target. If y^* is the desired property value and b_t is the best result collected so far, the remaining gap is $\Delta_t = |y^* - b_t|$. The relevant measure is how many experiments are needed for Δ_t to become sufficiently small, since a lower number indicates that the system approaches the target property with fewer laboratory tests. In reported studies, the optimization identified improved photo electrocatalytic conditions in nine cycles and reached $2.3\times$ higher activity than random sampling, while autonomous reaction optimization selected useful electrochemical conditions in ten or fewer iterations. Self-driving laboratories are

also presented as a way to reduce materials development timelines from 10–20 years to 1–2 years by linking automated laboratory work with models that decide the next experiment (Duo *et al.*, 2025; Su *et al.*, 2024).

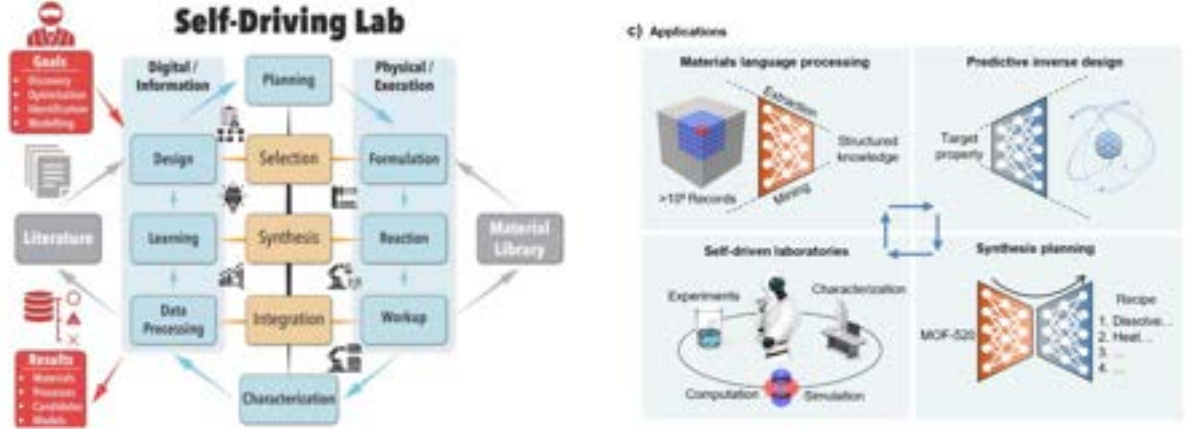


Figure 2.20: Self driving lab and agentic application in materials science. Sources (Duo *et al.*, 2025; Yuan *et al.*, 2025).

Another use case is a materials knowledge agent for formulation and synthesis planning, which connects technical documents and predictive tools to propose candidate materials to test. This is relevant for many industrial domains, such as polymers and batteries, where domain knowledge is distributed across papers and previous experimental results. The user specifies a target g , such as a formulation with higher heat resistance and lower cost, and the system retrieves a set of relevant sources $Z_K(g)$ from the available knowledge base. From the retrieved evidence, the agent builds a set of candidates $C = \{c_1, \dots, c_n\}$, where each c_i represents a possible experimental option and is evaluated through external tools, such as property prediction models, synthesis constraints, and process constraints. The selection step can be represented as $c^* = \arg \max_{c \in C} U(c)$, where $U(c)$ is a utility function that combines the expected value of a candidate with the cost and risk of testing it. For example, $U(c) = w_p \hat{p}(c) + w_u I(c) - w_c \text{Cost}(c) - w_r \text{Risk}(c)$, where $\hat{p}(c)$ is the predicted performance, $I(c)$ captures useful uncertainty when exploration is valuable, $\text{Cost}(c)$ measures material and process cost, and $\text{Risk}(c)$ includes process related risk. A research oriented workflow may give more weight to uncertainty and discovery, while an industrial workflow may give more weight to robustness, cost, and safety. As a result, the system reduces manual document search and supports expert review by grounding

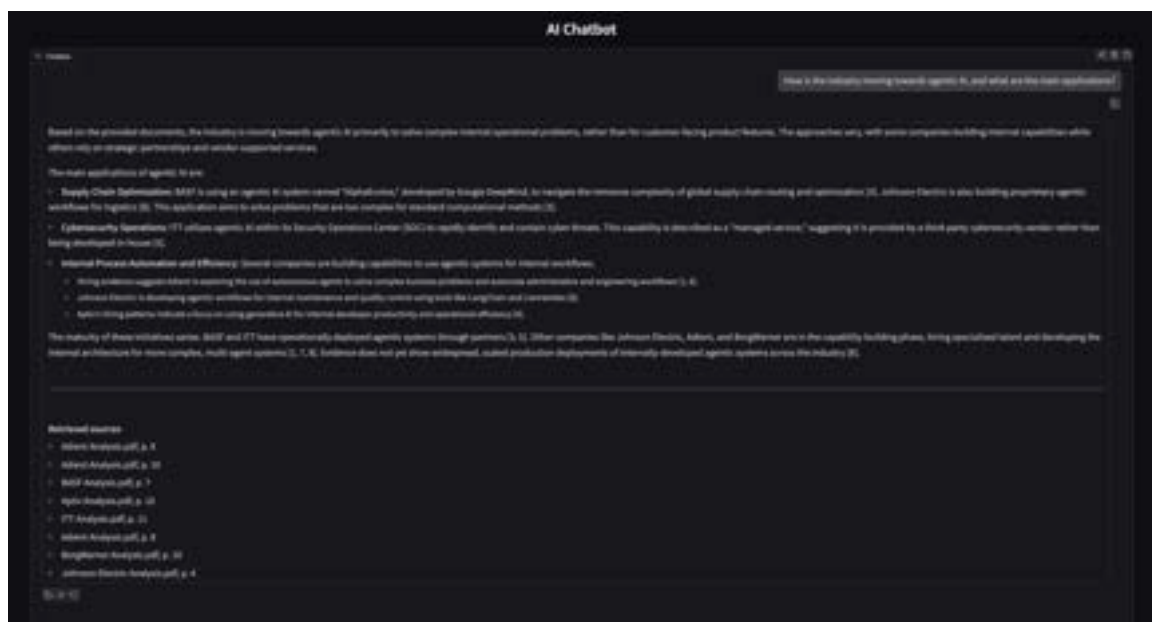
each recommendation in available experimental data and source based evidence. In the analyzed literature, automatic parsing reduced literature analysis time by more than $50\times$ compared to manual review and reported a large scale extraction pipeline that collected more than 300,000 material property records from about 130,000 abstracts in 60 hours (Yuan *et al.*, 2025).

3 Analysis

3.1 Study Design

The objective of this analysis is to understand how companies operating in manufacturing and chemistry are applying AI within their operations by using publicly available data. These firms were identified from Orbis by filtering for very large European firms in three NACE sectors, which were 22.11, manufacture of rubber tyres and tubes, 22.19, manufacture of other rubber products, and 29.32, manufacture of other parts and accessories for motor vehicles. These sectors were selected because they combine manufacturing, materials, and production intensive activities, where AI can appear in production, quality control, maintenance, product development, and digital operations. Each firm was then matched to its global corporate group, so the analysis covered worldwide hiring activity, avoiding the risk of overlooking AI roles located outside Europe. From this group, 16 firms that were matched with at least 2,000 job postings each were identified in TheirStack, including Michelin, Bridgestone, and Adient, which produced a final dataset of 91,544 hiring records from 2022 to the present. TheirStack is a source of historical online job postings collected from major hiring platforms, including Workday, Indeed, and LinkedIn. These records include job titles, required skills, locations, employers, and job descriptions, and provide signals about the internal competences that firms are trying to build. This approach was inspired by O’Kane’s use of online labor market data to measure AI demand through employer requested skills. However, these data are treated as hiring evidence instead of direct evidence of implemented AI projects because a job description can signal which internal competence a firm is trying to develop, but it may not show whether that competence has already been deployed (O’Kane, 2024). Since these insights are written as text, an LLM was used to extract AI related investment proxies and classify each posting according to the business area in which the technology was applied, the level of AI involvement in the role, and the main

AI technology mentioned. From the extracted fields, a hiring report was created for each company, summarizing the historical trend of AI roles, their seniority, the most frequent skills, the main technologies mentioned, and the business areas in which these technologies are applied. It also identified the capabilities that each firm seemed to be developing through hiring, treating them as operational priorities. However, job postings could omit technical details or describe roles in broad terms. For this reason, the hiring report was combined with annual reports and then expanded through a research agent, which searched for additional public sources such as interviews, press releases, investor communications, and conference material, and produced a final company report. A separate report was produced for each company and then evaluated to verify whether the agent extracted the information correctly without adding false or exaggerated claims. Finally, the reports were fed into a RAG chatbot that was used to test natural language queries about individual companies, sector trends, and specific technologies.



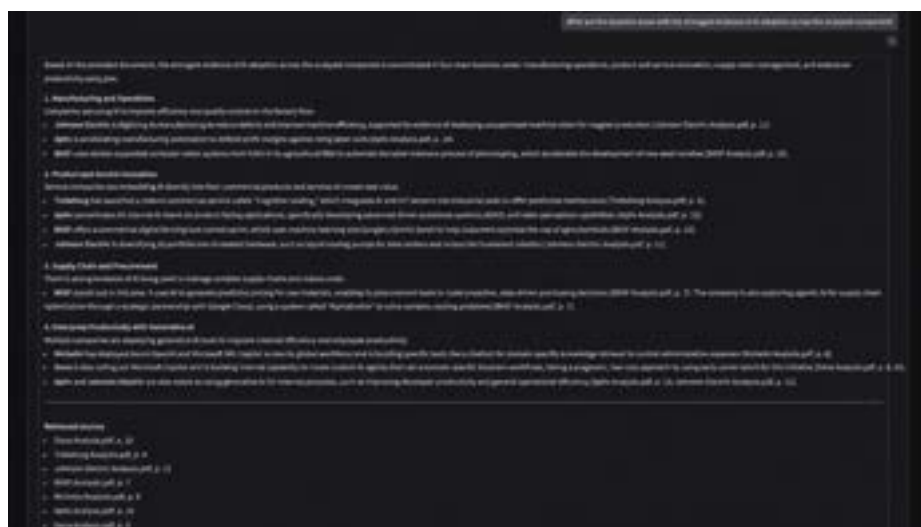
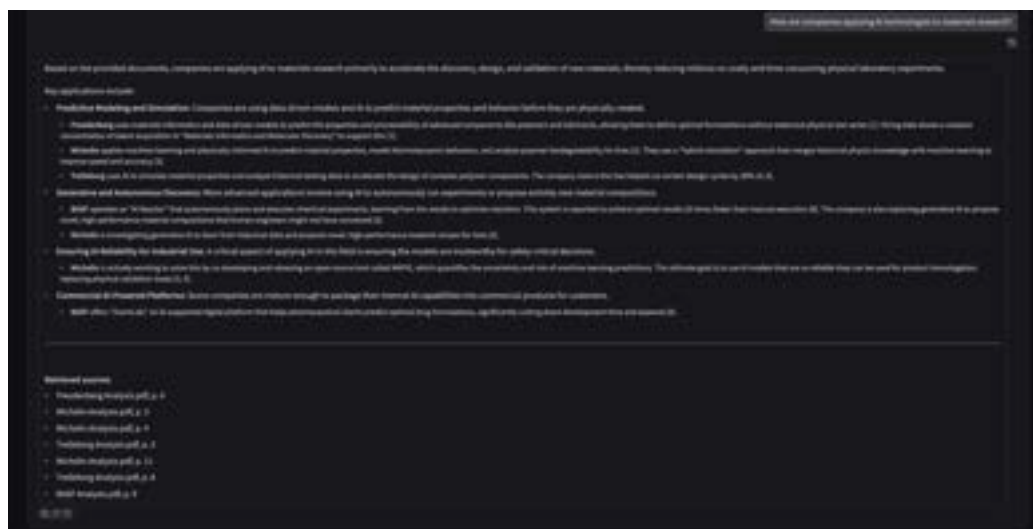
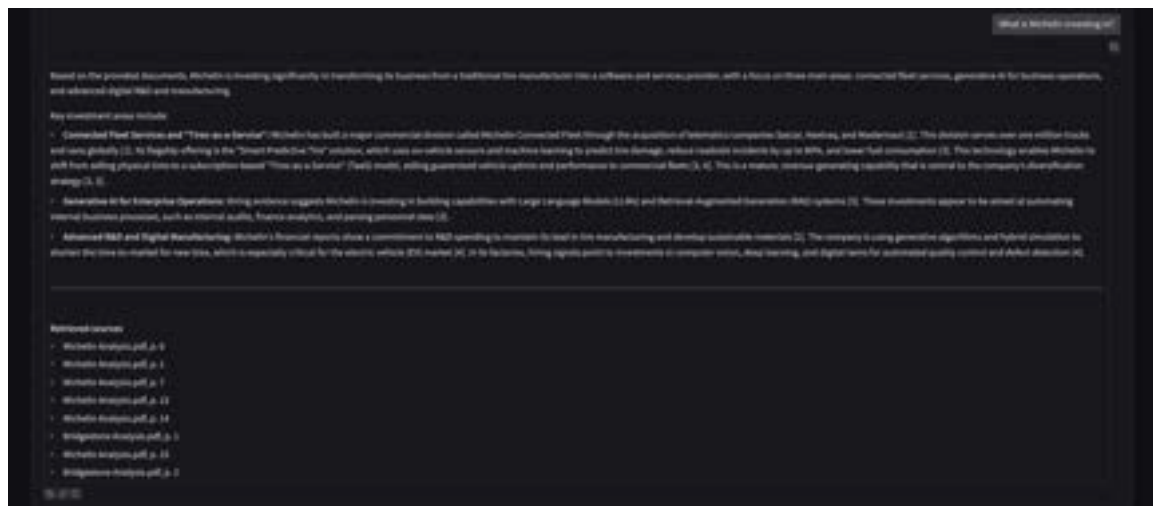


Figure 3.1: Examples of RAG chatbot answers based on final company reports.

3.2 Pipeline

For job posting classification, each row contained the job title, company name, seniority, country, posting date, and full job description. Gemini 3 Flash Preview processed each description and returned a structured JSON output, extracting five elements:

- `primary_category`, which describes the type of role being hired, such as business analytics, data engineering, cloud platforms, software engineering, enterprise systems, AI and ML modeling, LLM and GenAI applications, agentic AI systems, industrial automation, materials science R&D, non technical roles, or unclear cases.
- `business_area`, which identifies where the technology is applied inside the firm, such as manufacturing operations, materials research, IT and data, supply chain, customer offerings, sales and marketing, or corporate functions.
- `technical_relevance_score`, which measures the role's connection to AI. A score of 2 means that the person is hired to build, develop, or apply AI directly in a product, process, model, or internal system. A score of 1 means that the person is expected to know or use AI related tools, but is not hired to create an AI capability. A score of 0 means that the role is not meaningfully related to AI.
- `technical_signals`, which lists the explicit tools, methods, platforms, technical systems, and skills mentioned in the posting.
- `investment_signal`, which is generated only when the score is 2. It summarizes the AI capability that the firm is trying to build and the business use case behind it.

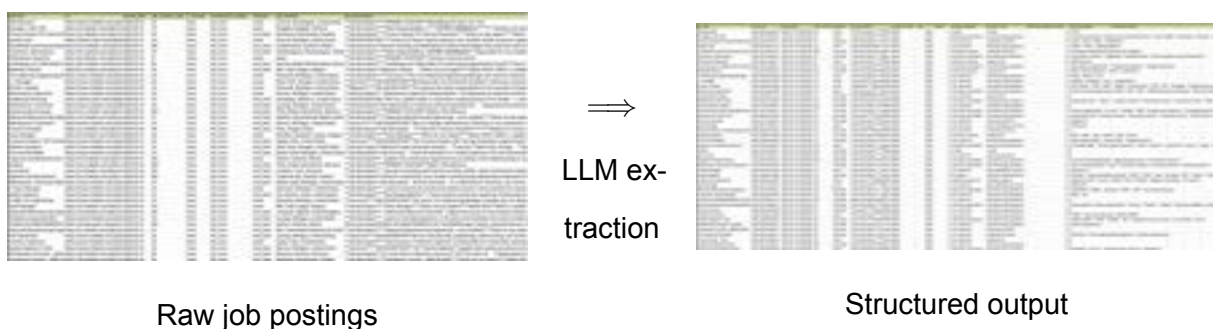
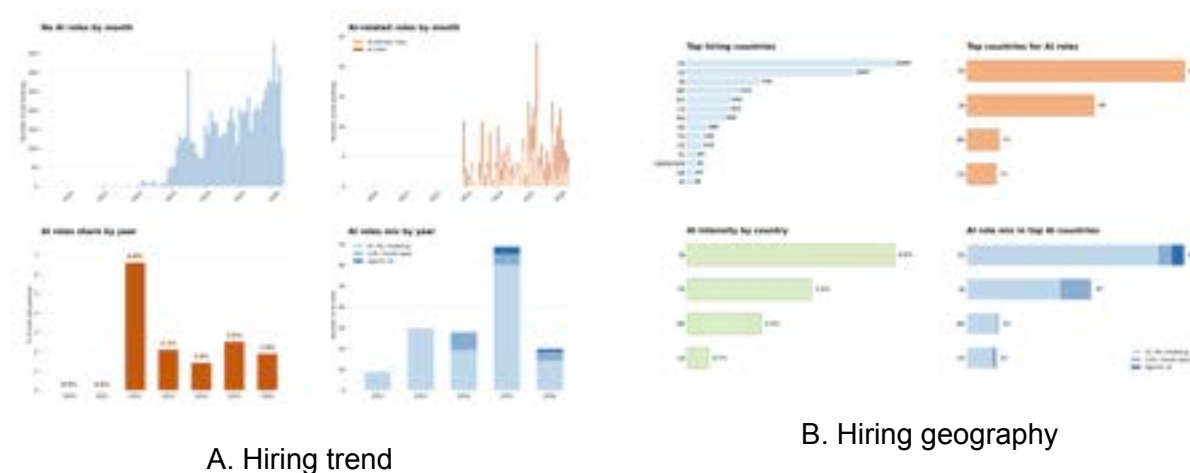
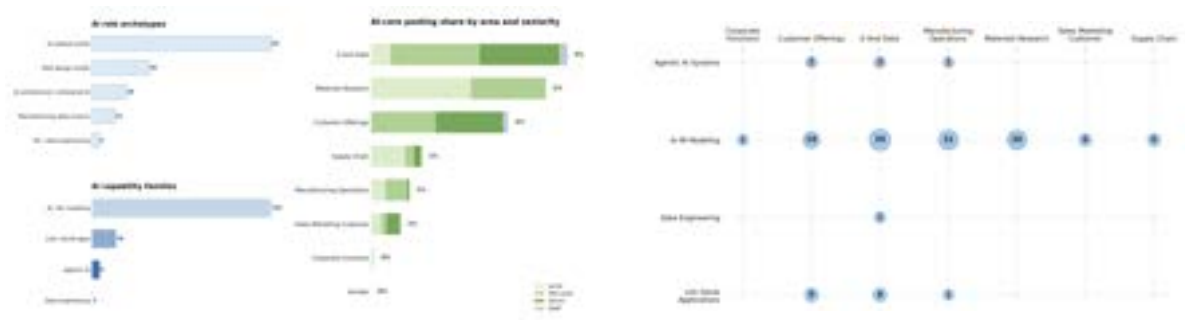


Figure 3.2: Transformation of job posting descriptions into structured fields through LLM based extraction.

After the classification, the processed dataset was used to create one hiring report for each company. The report combined four views of the data, each one designed to summarize a different part of the recruitment information:

- **Hiring trend:** This shows how hiring changed over time, separating roles with no AI signal, AI adjacent roles, and AI core roles. It was used to understand whether AI related hiring was stable, increasing, or concentrated in specific years.
- **Hiring geography:** This compares the countries with the largest overall number of postings with the countries where AI core postings appeared. It was used to distinguish general recruitment activity from AI specific recruitment activity.
- **Staffing and seniority:** This summarizes the type of AI profiles being hired and their seniority, showing whether AI roles are concentrated in junior, mid level, senior, or staff positions, and in which business areas they appear.
- **Breadth across capability pockets:** This maps AI core roles across primary_category and business_area. It was used to identify where AI hiring is concentrated by combining the type of capability with the area where it is applied.



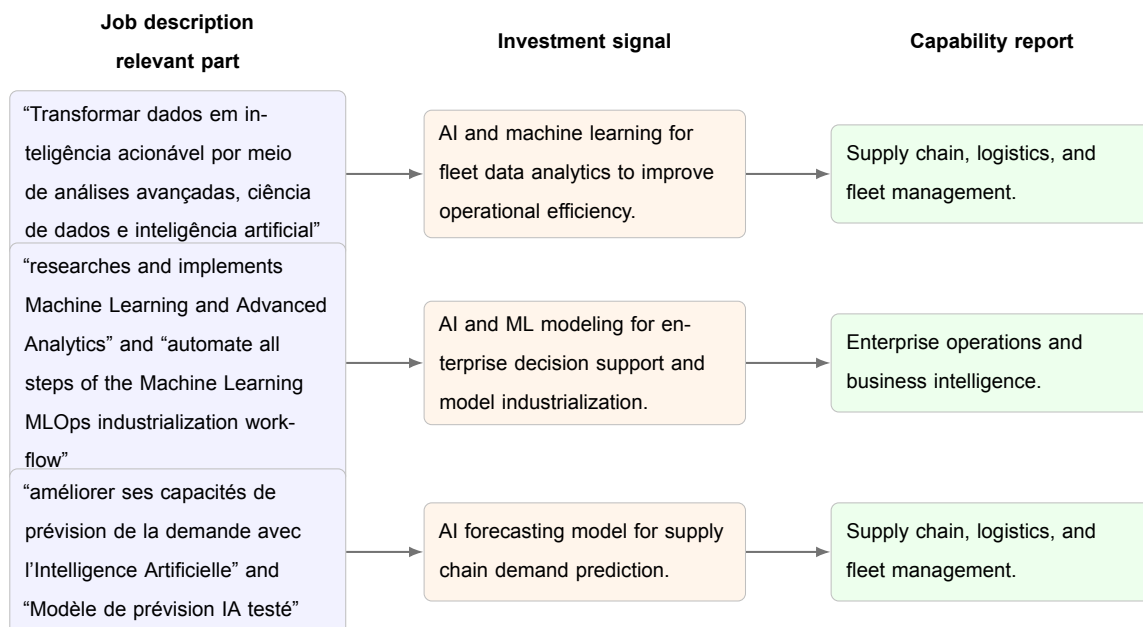


C. Staffing and seniority

D. Breadth across capability pockets

Figure 3.3: Views used in the hiring report to summarize AI related recruitment patterns.

The `investment_signal` column was then used to create a capability report for each company. Since this field was generated only for score 2 postings, it captured the cases where AI was directly used to build or apply a capability. For each company, all these signals were collected and passed to another LLM call, which grouped them into broader capability areas, such as fleet analytics, enterprise decision support, demand forecasting, materials research, or industrial automation. These were then used as the inputs for the research agent, which searched public sources to validate, expand, or contextualize the signals found in the job postings.



A. From job descriptions to capability signals

Industrial Process Optimization and Manufacturing Automation
The company is investing in deep learning, computer vision, and agents to enhance manufacturing process control and equipment health monitoring. These technologies are applied to industrial data for equipment recovery, defect detection, and quality inspection, as well as IoT signal processing for real-time equipment monitoring. The objective is to optimize production efficiency, reduce unplanned costs, and enable predictive maintenance through self-learning models and path optimization.

Material Science and Product R&D Innovation
Investment in physical model-machine learning, graph neural networks, and generative AI to accelerate material design and characterization. These capabilities are used for predicting material properties, thermodynamic behavior, and process feasibility, specifically targeting bio-performance optimization and nanoparticle production modeling. The goal is to reduce experimental costs, accelerate R&D cycles, and ensure product compliance in critical markets.

Predictive Maintenance and Product Performance Analytics
The company is building predictive modeling and hybrid AI systems capabilities to monitor and enhance product quality, performance for fleet and industrial equipment. Applications include machine health monitoring, risk score prediction, and predictive maintenance planning for mining and manufacturing operations. These capabilities aimed to reduce equipment downtime, optimize asset utilization, and improve safety and sustainability.

Customer Engagement and Personalized AI
There is a significant focus on large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and generative AI to develop customer-facing AI products. These technologies power chatbots and conversational interfaces designed to enhance user engagement, improve digital service offerings, and streamline customer journey interactions across various digital channels.

Supply Chain, Logistics, and Fleet Management
AI and machine learning are being leveraged to demand forecasting, procurement efficiency, and fleet data analysis. Supply chain optimization agents for the connected fleet management, route optimization, scheduling, and capacity use prediction. The objective is to optimize supply chain planning, improve cost safety, and enhance the operational efficiency of logistics and fleet services.

Marketing Intelligence and Consumer Insights
The company is focusing on predictive modeling and generation AI for marketing automation, customer segmentation, and churn modeling. These tools are used to analyze consumer behavior, optimize campaign timing, and drive strategic product portfolio decisions. The primary goal is to maximize long-term customer value, reduce churn, and increase revenue through data-driven marketing strategies.

Enterprise Operations and Business Intelligence
Investment in AI and machine learning is being translated into internal business process automation, including internal audit, finance, and personnel data analysis. Applications include the generation of financial statements, loan rules, and automation of operations. These capabilities aim to drive business transformation, enhance operational excellence, and support data-driven decision-making across the enterprise.

AI Infrastructure and MLOps
To support its AI initiatives, the company is investing in MLOps and scalable data infrastructure, particularly on the cloud-based platform. This includes building end-to-end ML pipelines, AI capabilities for model lifecycle, and optimization for high-throughput, low-latency use cases. The objective is to provide the reliable, secure, and scalable deployment of AI models into production environments.

AI Operations and Cybersecurity
The company is investing generative AI and machine learning in IT operations (GenIT for SRE) and cyber threat detection. These investments are focused on improving system reliability, enhancing real-time response capabilities to security threats, and reducing the operational costs of maintaining IT infrastructure.

B. Capability report output

Figure 3.4: Example of how investment signals extracted from job postings were grouped into a company capability report.

After the capability report was created, it was combined with the annual report of the same company and a Google research agent, deep-research-preview-04-2026, was used to expand the evidence base. The annual report provided more formal and quantitative information, such as R&D expenses, capital expenditure, intangible assets, capitalized software when disclosed, acquisitions, partnerships, and notes on digital or AI related initiatives. Instead, job postings were used to provide more granular signals about the capabilities that the company was trying to build through recruitment. The research agent then searched public sources such as press releases, company pages, and leadership interviews to extend these signals and clarify what the firm was actually building, using, or scaling. The final result was one company report structured around four sections:

- **Quantitative investment baseline:** This used the annual report to collect financial and strategic items related to technology investment, including R&D expenses, capital expenditure, intangible assets, capitalized software, acquisitions, partnerships, and disclosed digital initiatives. These figures were considered technology evidence only when linked to software, digital systems, or AI related activity.

- **Capability-by-capability analysis, including GenAI and agentic systems:** This analyzed the main capability areas identified from the hiring report and public evidence. For each area, it described what the company was building or using, where it was applied, which problem it addressed, and whether the evidence pointed to capability building, operational use, or commercial deployment.
- **Talent, teams, and operating model:** This used the hiring data to describe where AI, automation, and digital roles appeared inside the firm. It focused on business areas, countries, seniority, and whether the evidence pointed to internal hiring, external partnerships, vendor platforms, or acquisitions.
- **What the evidence implies for investment priorities and competitor monitoring:** This summarized the main AI and digital priorities that emerged. It identified which capabilities seemed more relevant, what remained uncertain, and which signals a competitor should monitor over time.

After each final company report was generated, it was evaluated with a separate benchmark that checked public source claims against their cited evidence and flagged unsupported, non verifiable, or potentially overstated statements. The evaluated reports were then embedded with multilingual-e5-small and indexed in a RAG chatbot, where they served as the knowledge base for testing natural language queries about individual companies, sector trends, and specific technologies. See Figure 3.1.

3.3 Evaluation

Since the final reports were used as the knowledge base of the RAG chatbot, their reliability directly affected the quality of the generated answers. The evaluation therefore assessed whether the factual claims in each report were supported by the evidence cited by the research agent. For this analysis, claims linked to public URLs were used, since their citations made it possible to compare each statement with the external source used by the agent. For each report, the evaluation first extracted the bibliography and created a citation map, linking each reference number to its title, publisher, and URL when available. The text was then scanned to extract every factual and checkable claim that cited at least one public URL. If one sentence contained more than one factual statement, it was split into separate claims, so that each claim could be verified independently. Each URL was then processed with Crawl4AI, which converted the webpage or PDF into plain readable text. Based on this extraction, each source was classified as accessible if enough readable text was recovered, `not_readable` if the URL was reached but the text was not usable, and inaccessible if the URL could not be retrieved. The recovered text was split into shorter overlapping excerpts, meaning small passages taken from the original public source. A TF-IDF similarity search then selected the excerpts most similar to the claim, so that the verification step used only the most relevant parts of the source. At this point, `gemini-3.1-pro-preview` was used as a judge to compare each claim with the selected excerpts and assign one of four labels:

- `supported`: the extracted passages contained the information needed to justify the statement.
- `needs_review`: the extracted passages supported part of the statement, while other parts required narrower wording, additional evidence, or manual review.
- `unsupported`: the extracted passages did not contain the evidence needed to justify the statement, or referred to a different point.
- `not_verifiable`: the statement could not be checked because the URL could not

be retrieved, the page could not be parsed, or the recovered text was too limited.

To analyze the results, both these labels and the claim and source counts were used to compute metrics for each company report:

- **URL coverage:** the share of extracted claims that had at least one public URL citation and could enter the benchmark, computed as $\frac{\text{claims checked with public URLs}}{\text{total extracted claims}}$.
- **Grounding accuracy:** a weighted measure of how well the claims were supported by the cited evidence, computed as $\frac{\text{supported} + 0.5 \times \text{needs review}}{\text{supported} + \text{needs review} + \text{unsupported}}$.
- **Strict support rate:** the share of verifiable claims that were fully supported by the cited evidence, computed as $\frac{\text{supported}}{\text{supported} + \text{needs review} + \text{unsupported}}$.
- **Review burden rate:** the share of verifiable claims that required manual review or softer wording, computed as $\frac{\text{needs review}}{\text{supported} + \text{needs review} + \text{unsupported}}$.
- **Unsupported rate:** the share of verifiable claims that were not supported by the cited evidence, computed as $\frac{\text{unsupported}}{\text{supported} + \text{needs review} + \text{unsupported}}$.
- **Verifiability rate:** the share of public source claims that could actually be judged after retrieving the sources, computed as $\frac{\text{supported} + \text{needs review} + \text{unsupported}}{\text{supported} + \text{needs review} + \text{unsupported} + \text{not verifiable}}$.

Therefore, the metrics were designed to distinguish claim quality from source availability issues, so that low scores could be traced either to report errors, wording issues, or source access problems.

Company	Claims	GA	SSR	RBR	UR	VR
BASF	21	85.7%	80.9%	4.8%	14.3%	100.0%
Trelleborg	21	66.7%	52.4%	47.6%	0.0%	100.0%
BorgWarner	18	42.3%	23.1%	38.5%	38.5%	72.2%
Dana	68	76.5%	65.3%	22.4%	12.2%	72.1%

Table 3.1: Examples of benchmark outcomes.

For instance, BASF could be used with limited revision, because most claims were supported and verifiable. Trelleborg needed wording edits, since the issue was mainly

review burden. BorgWarner required direct correction, because several claims were not justified by the extracted source passages. Dana shows a solid result, with most verifiable claims grounded in the extracted evidence and only a limited share requiring correction.



Figure 3.5: Overview of the evaluation workflow.

Overall, the batch results show that most verifiable claims were grounded in the extracted evidence, but not all reports reached the same reliability level. Indeed, a relevant share of claims needed more precise wording, while another part could not be checked because the cited source could not be retrieved or parsed automatically.

Metric	Batch result
Public URL claims	392
Judged claims	292
Supported claims	174
Claims needing review	75
Unsupported claims	43
Not verifiable claims	100
Grounding accuracy	72.4%
Strict support rate	59.6%
Review burden rate	25.7%
Unsupported rate	14.7%
Verifiability rate	74.5%

Table 3.2: Benchmark results across the full batch of company reports.

A possible extension would be to generate final reports for all companies identified

through Orbis in the selected sectors and then run the same benchmark on them. A further extension would be to add an iterative correction step in which an LLM uses the benchmark output to revise each report and repeat the evaluation until the main grounding metrics improve.

4 Conclusions

This thesis analyzed how Artificial Intelligence is being applied in manufacturing and chemistry, starting from the applications already described in the technical literature and then moving to company evidence collected from public sources. Across the review, the main change was the passage from narrow models built for specific prediction tasks to systems that can connect technical language, external evidence, and software tools. This is relevant in manufacturing and chemistry because plant data, laboratory records, manuals, technical reports, and business information are often handled separately. Newer systems based on LLMs, RAG, and agentic AI can therefore support the search, coordination, and use of this information, while researchers, engineers, and operators remain responsible for judgment and validation. At the same time, the evidence reviewed in the thesis suggests that many examples show strong technical potential, but they often remain prototypes, research workflows, or early applications. This was also reflected in the empirical part, where the evaluated company reports were read directly and then queried through the RAG chatbot to compare companies, technologies, and capability areas. More specifically, the company analysis found clearer evidence for more established classical AI uses, such as predictive modelling and analytics, while newer technologies, especially agentic AI, were more limited and often pointed to current capability development.

The empirical part of the thesis aimed to reconstruct what could be observed from the outside, with company reports acting as the link between fragmented public evidence and the RAG chatbot. Since these reports were used as the chatbot knowledge base, their quality directly affected the value of the experiment. The evaluation showed that most verifiable claims were grounded in the extracted evidence, while also identifying claims that needed more precise wording, removal, or additional checks because the cited source did not support them or could not be retrieved automatically. The main limitation is that this work remains based on external evidence, so it cannot

fully observe internal deployment, production maturity, or actual spending unless these are explicitly disclosed. For this reason, the results should be interpreted as signals of what companies could be developing and can still be useful for competitors and sector experts, since they help connect public evidence with possible applications of the technology.

In terms of future work, the first step would be to apply the same procedure to all companies identified through Orbis in the selected sectors, showing whether the patterns observed in the current sample also appear across a larger group of firms. The pipeline could then be redesigned as a more agentic workflow, because the current version executes each step once and in a fixed order, so weak evidence or unsupported claims are only detected at the evaluation stage. A future version could use the benchmark results earlier in the process, so that the system can search again when a source is weak, rewrite claims that are too broad, remove unsupported statements, and repeat the evaluation after correction. The same principle could also be applied to the RAG chatbot, where a judge could compare each generated answer with the retrieved passages before returning it to the user.

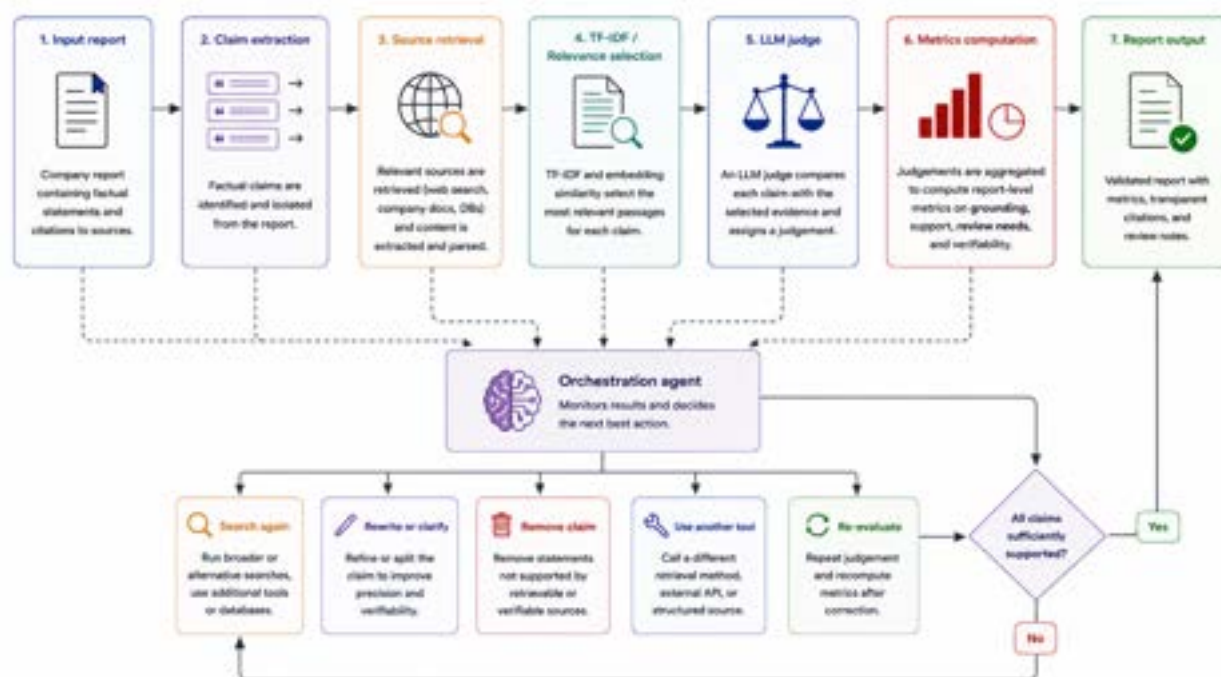


Figure 4.1: Proposed agentic extension of the evaluation workflow.

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